

Intelligent Voice Recognition & Automation Assistant

Test Cases:

1. Module: Voice Recognition :

Functional Requirements Covered

- Capture voice input through microphone
- Convert speech into text
- Continuous voice interaction
- Background noise filtering
- Handle unrecognized speech
- Forward recognized text to Command Processing

TC ID	Test Case Description	Expected Result	Result (Pass/Fail)
TC01	Open the Voice Recognition module.	Module opens successfully.	
TC02	Speak a simple command into the microphone.	Voice is captured successfully.	
TC03	Speak "Open Notepad".	Speech is converted into text correctly.	
TC04	Speak "Open Calculator".	Correct text is generated.	
TC05	Speak a long sentence.	Complete speech is converted into text.	
TC06	Speak slowly.	Speech is recognized correctly.	

TC07	Speak quickly.	Speech is recognized correctly.	
TC08	Speak in a normal voice.	Voice is recognized successfully.	
TC09	Speak in a low voice.	System recognizes the command or asks to repeat.	
TC10	Speak loudly.	Voice is recognized correctly.	
TC11	Stay silent after the microphone starts listening.	"No speech detected" message is displayed.	
TC12	Give an unsupported voice command.	"Command not recognized" message is displayed.	
TC13	Speak random words.	Invalid command message is displayed.	
TC14	Speak with fan noise in the background.	Background noise is filtered and command is recognized.	
TC15	Speak while music is playing.	System recognizes the command or asks to repeat.	
TC16	Speak while TV sound is playing.	System handles background noise properly.	
TC17	Speak while another person is talking.	Primary voice is recognized or retry is requested.	
TC18	Use the built-in laptop microphone.	Voice is captured successfully.	
TC19	Use a USB microphone.	Voice is captured successfully.	

TC20	Use a Bluetooth microphone.	Voice is captured successfully.	
TC21	Disconnect the microphone while listening.	Error message is displayed.	
TC22	Disable microphone permission in Windows.	Permission error is displayed.	
TC23	Reconnect the microphone and speak again.	Voice Recognition resumes successfully.	
TC24	Speak numbers in a command.	Numbers are converted correctly.	
TC25	Speak application names.	Application names are recognized correctly.	
TC26	Speak file names.	File names are recognized correctly.	
TC27	Give two commands one after another.	Both commands are recognized separately.	
TC28	Pause while speaking.	System waits briefly and processes the complete command.	
TC29	Repeat the same command several times.	Every command is recognized correctly.	
TC30	Stop speaking before completing the command.	Incomplete command is ignored or retry is requested.	
TC31	Keep speaking continuously for one minute.	Continuous speech is processed successfully.	
TC32	Start Voice Recognition immediately after launching NOVA.	Voice Recognition starts successfully.	
TC33	Minimize NOVA while Voice Recognition is active.	Voice Recognition continues if supported.	

TC34	Restore NOVA and continue speaking.	Recognition continues normally.	
TC35	Restart the Voice Recognition module.	Module restarts successfully.	
TC36	Give a voice command after restarting the module.	Command is recognized successfully.	
TC37	Verify recognized text is forwarded to Command Processing.	Text is forwarded successfully.	
TC38	Verify empty speech is not forwarded.	No command is sent to Command Processing.	
TC39	Give 20 voice commands continuously.	All commands are recognized successfully.	
TC40	Run Voice Recognition continuously for 30 minutes.	Module remains stable without crashing.	
TC41	Check CPU usage during continuous listening.	CPU usage remains within acceptable limits.	
TC42	Check memory usage during continuous listening.	Memory usage remains stable.	
TC43	Verify response time after speaking a command.	Text is generated within acceptable time.	
TC44	Verify recognized command accuracy.	Recognized text matches the spoken command.	
TC45	Verify Voice Recognition after Windows restart.	Module works correctly.	
TC46	Verify Voice Recognition after application restart.	Module initializes successfully.	
TC47	Verify Voice Recognition log is created (if logging is enabled).	Recognition event is logged successfully.	

TC48	Close the Voice Recognition module while listening.	Module closes safely without errors.	
TC49	Exit NOVA after using Voice Recognition.	Application exits successfully.	
TC50	Verify Voice Recognition resources are released after exit.	Microphone and related resources are released properly.	

2. Module: Registry-Based Application Launcher :

Functional Requirements

- Search the Windows Registry for installed applications.
- Retrieve the executable path.
- Launch applications without hard-coded paths.
- Display message if application is not found.
- Support Windows and third-party applications.
- Ensure compatibility across Windows systems.

TC ID	Test Case Description	Expected Result	Result (Pass/Fail)
TC01	Start the Registry Launcher module.	Module starts successfully.	
TC02	Search for Notepad in the Windows Registry.	Notepad entry is found.	
TC03	Search for Calculator in the Windows Registry.	Calculator entry is found.	
TC04	Search for Paint in the Windows Registry.	Paint entry is found.	

TC05	Search for Microsoft Word (if installed).	Registry entry is found.	
TC06	Search for Google Chrome (if installed).	Registry entry is found.	
TC07	Search for Visual Studio Code (if installed).	Registry entry is found.	
TC08	Search for an application using its full name.	Correct application is found.	
TC09	Search for an application using a partial name.	Matching application is displayed.	
TC10	Search for an application that is not installed.	"Application not found" message is displayed.	
TC11	Search with an empty application name.	Validation message is displayed.	
TC12	Search using only spaces.	Search is not performed.	
TC13	Search using uppercase letters.	Application is found successfully.	
TC14	Search using lowercase letters.	Application is found successfully.	
TC15	Search using mixed-case letters.	Search works correctly.	
TC16	Retrieve executable path of Notepad.	Correct executable path is returned.	
TC17	Retrieve executable path of Calculator.	Correct executable path is returned.	
TC18	Verify executable path exists.	Valid executable path is found.	

TC19	Verify executable path is not empty.	Path is displayed correctly.	
TC20	Launch Notepad using the registry path.	Notepad opens successfully.	
TC21	Launch Calculator using the registry path.	Calculator opens successfully.	
TC22	Launch Paint using the registry path.	Paint opens successfully.	
TC23	Launch Chrome using the registry path.	Chrome opens successfully.	
TC24	Launch VS Code using the registry path.	VS Code opens successfully.	
TC25	Launch an application that is not installed.	Appropriate error message is displayed.	
TC26	Attempt to launch an application with an invalid registry entry.	Error is handled properly.	
TC27	Launch multiple applications one by one.	Each application launches successfully.	
TC28	Launch the same application multiple times.	Application launches successfully each time.	
TC29	Verify application launches without hard-coded paths.	Application launches through registry only.	
TC30	Verify registry values are read correctly.	Correct registry information is retrieved.	
TC31	Verify registry access permission.	Registry is accessed successfully.	

TC32	Registry key is missing.	Proper error message is displayed.	
TC33	Registry value is empty.	Error is displayed.	
TC34	Registry contains invalid executable path.	Launch fails with appropriate message.	
TC35	Verify 64-bit application registry lookup.	Application is found correctly.	
TC36	Verify 32-bit application registry lookup.	Application is found correctly.	
TC37	Verify compatibility on Windows 10.	Module works correctly.	
TC38	Verify compatibility on Windows 11.	Module works correctly.	
TC39	Restart Registry Launcher module.	Module restarts successfully.	
TC40	Search immediately after module restart.	Search works correctly.	
TC41	Perform 50 consecutive searches.	All searches complete successfully.	
TC42	Perform continuous launches for different applications.	No crash occurs.	
TC43	Verify response time while searching applications.	Search completes within acceptable time.	
TC44	Verify memory usage during registry search.	Memory usage remains stable.	

TC45	Verify CPU usage during registry search.	CPU usage remains acceptable.	
TC46	Close Registry Launcher while searching.	Module closes safely.	
TC47	Restart application after abnormal termination.	Registry Launcher works normally.	
TC48	Verify search results after Windows restart.	Registry search works correctly.	
TC49	Verify logs are generated for registry searches.	Search activity is logged successfully.	
TC50	Exit Registry Launcher module.	Module exits without errors.	

3. Module: Windows Startup Integration :

Functional Requirements

- Register NOVA in Windows Startup Registry.
- Launch assistant automatically after user login.
- Retrieve current Windows username.
- Display personalized welcome message.
- Run assistant in background.
- Enable/Disable auto-start.

TC ID	Test Case Description	Expected Result	Result (Pass/Fail)
TC01	Start the Windows Startup module.	Module starts successfully.	
TC02	Enable Auto Start for NOVA.	Startup entry is added successfully.	
TC03	Disable Auto Start for NOVA.	Startup entry is removed successfully.	
TC04	Verify Startup Registry entry after enabling Auto Start.	Registry entry exists.	
TC05	Verify Startup Registry entry after disabling Auto Start.	Registry entry is removed.	
TC06	Restart Windows after enabling Auto Start.	NOVA launches automatically.	
TC07	Log in to Windows after enabling Auto Start.	NOVA starts automatically.	
TC08	Verify NOVA starts only once after login.	Only one instance runs.	
TC09	Enable Auto Start multiple times.	Duplicate registry entries are not created.	
TC10	Disable Auto Start multiple times.	No error occurs.	
TC11	Retrieve current Windows username.	Correct username is displayed.	
TC12	Verify username after another user logs in.	Correct logged-in username is displayed.	
TC13	Username contains spaces.	Username is displayed correctly.	

TC14	Username contains numbers.	Username is displayed correctly.	
TC15	Username contains special characters.	Username is displayed correctly.	
TC16	Display welcome message after login.	Personalized welcome message appears.	
TC17	Verify welcome message contains username.	Correct username is displayed.	
TC18	Verify welcome message is shown only once.	Welcome message appears once per login.	
TC19	Verify background process starts automatically.	NOVA runs in background.	
TC20	Verify background process after Windows restart.	Background process starts successfully.	
TC21	Minimize NOVA after startup.	Application continues running in background.	
TC22	Close NOVA manually after startup.	Application exits successfully.	
TC23	Enable Auto Start without administrator privileges.	Appropriate error or warning is displayed.	
TC24	Verify startup when Registry access is denied.	Error message is displayed.	
TC25	Verify startup when Startup Registry key is missing.	Module handles the error properly.	

TC26	Enable Auto Start after disabling it.	Auto Start works correctly.	
TC27	Disable Auto Start after enabling it.	Auto Start is disabled successfully.	
TC28	Verify Auto Start status in Settings.	Correct status is displayed.	
TC29	Restart NOVA after enabling Auto Start.	Auto Start setting remains saved.	
TC30	Restart NOVA after disabling Auto Start.	Auto Start remains disabled.	
TC31	Verify Startup module on Windows 10.	Module works correctly.	
TC32	Verify Startup module on Windows 11.	Module works correctly.	
TC33	Verify startup after Windows Update.	NOVA starts successfully.	
TC34	Verify startup after system sleep.	Background service resumes normally.	
TC35	Verify startup after system hibernate.	Background service resumes correctly.	
TC36	Verify startup after unexpected shutdown.	NOVA starts normally after login.	
TC37	Verify CPU usage during startup.	CPU usage remains acceptable.	
TC38	Verify memory usage during startup.	Memory usage remains stable.	
TC39	Measure startup time of NOVA.	Application starts within expected time.	
TC40	Verify startup log creation.	Startup activity is logged.	

TC41	Verify username retrieval without internet.	Username is displayed correctly.	
TC42	Verify startup with slow system boot.	NOVA launches successfully.	
TC43	Verify Auto Start after changing Windows password.	Auto Start continues working.	
TC44	Verify Auto Start after changing Windows username.	Updated username is displayed.	
TC45	Verify Auto Start when another startup application is running.	NOVA starts successfully.	
TC46	Verify startup after reinstalling NOVA.	Auto Start functions correctly.	
TC47	Verify registry values remain unchanged after restart.	Startup Registry values are correct.	
TC48	Close Startup module during execution.	Module closes safely.	
TC49	Restart Startup module.	Module starts successfully.	
TC50	Exit Startup module.	Module exits without errors.	

4. Module: File & Folder Management :

Functional Requirements

- Search files and folders
- Open files and folders
- Create new folders
- Rename files and folders
- Delete files with user confirmation
- Navigate to requested directories

TC ID	Test Case Description	Expected Result	Result (Pass/Fail)
TC01	Open the File & Folder Management module.	Module opens successfully.	
TC02	Search for an existing file by name.	Correct file is displayed.	
TC03	Search for an existing folder by name.	Correct folder is displayed.	
TC04	Search for a file that does not exist.	"File not found" message is displayed.	
TC05	Search for a folder that does not exist.	"Folder not found" message is displayed.	
TC06	Search using a partial file name.	Matching files are displayed.	
TC07	Search using a partial folder name.	Matching folders are displayed.	
TC08	Search with an empty input.	Validation message is displayed.	
TC09	Search using only spaces.	Search is not performed.	

TC10	Search for files with special characters in the name.	Matching files are displayed correctly.	
TC11	Open an existing text file.	File opens successfully.	
TC12	Open an existing PDF file.	PDF opens with the default application.	
TC13	Open an image file.	Image opens successfully.	
TC14	Open an existing folder.	Folder opens in File Explorer.	
TC15	Try to open a deleted file.	Error message is displayed.	
TC16	Try to open a deleted folder.	Error message is displayed.	
TC17	Create a new folder with a valid name.	Folder is created successfully.	
TC18	Create a folder inside another folder.	Folder is created successfully.	
TC19	Create multiple folders one after another.	All folders are created successfully.	
TC20	Create a folder with an existing name.	Warning message is displayed.	
TC21	Create a folder using invalid characters.	Folder is not created and validation message is displayed.	
TC22	Rename an existing file.	File name is updated successfully.	
TC23	Rename an existing folder.	Folder name is updated successfully.	
TC24	Rename a file to an existing file name.	Rename operation is prevented.	
TC25	Rename a folder to an existing folder name.	Rename operation is prevented.	

TC26	Rename a file using invalid characters.	Validation message is displayed.	
TC27	Rename a folder using invalid characters.	Validation message is displayed.	
TC28	Delete an existing file and select Yes in the confirmation dialog.	File is deleted successfully.	
TC29	Delete an existing file and select No in the confirmation dialog.	File is not deleted.	
TC30	Delete a folder after confirmation.	Folder is deleted successfully.	
TC31	Cancel folder deletion.	Folder remains unchanged.	
TC32	Try to delete a file that does not exist.	Appropriate error message is displayed.	
TC33	Try to delete a folder that does not exist.	Appropriate error message is displayed.	
TC34	Navigate to the Documents directory.	Documents folder opens successfully.	
TC35	Navigate to the Downloads directory.	Downloads folder opens successfully.	
TC36	Navigate to the Desktop directory.	Desktop folder opens successfully.	
TC37	Navigate to a custom folder path.	Requested folder opens successfully.	
TC38	Navigate to an invalid folder path.	Error message is displayed.	
TC39	Navigate using a voice command.	Correct folder opens successfully.	
TC40	Perform multiple search operations continuously.	All searches complete successfully.	

TC41	Create and rename a folder immediately.	Both operations complete successfully.	
TC42	Search for a newly created folder.	Folder appears in search results.	
TC43	Delete the newly created folder.	Folder is removed successfully.	
TC44	Perform file operations on a read-only file.	Permission error is displayed.	
TC45	Verify application remains stable during multiple file operations.	No crash occurs.	
TC46	Verify file operation logs are generated.	Log entries are created successfully.	
TC47	Restart the File & Folder Management module.	Module starts successfully.	
TC48	Perform file operations after restarting the application.	All operations work correctly.	
TC49	Close the File & Folder Management module.	Module closes successfully.	
TC50	Exit the application after file operations.	Application exits without errors.	

5. Module: Application Launcher :

Functional Requirements

- Launch Windows applications.
- Launch third-party applications.
- Receive application name from Command Processing.
- Open applications using the Registry Launcher.
- Display an error if the application cannot be launched.
- Handle multiple launch requests.

TC ID	Test Case Description	Expected Result	Result (Pass/Fail)
TC01	Open the Application Launcher module.	Module starts successfully.	
TC02	Launch Notepad.	Notepad opens successfully.	
TC03	Launch Calculator.	Calculator opens successfully.	
TC04	Launch Paint.	Paint opens successfully.	
TC05	Launch Command Prompt.	Command Prompt opens successfully.	
TC06	Launch File Explorer.	File Explorer opens successfully.	
TC07	Launch Microsoft Edge.	Edge browser opens successfully.	
TC08	Launch Google Chrome (if installed).	Chrome opens successfully.	
TC09	Launch Visual Studio Code (if installed).	VS Code opens successfully.	

TC10	Launch Microsoft Word (if installed).	Word opens successfully.	
TC11	Launch Microsoft Excel (if installed).	Excel opens successfully.	
TC12	Launch an application using its full name.	Correct application opens.	
TC13	Launch an application using a partial name.	Closest matching application opens.	
TC14	Launch an application with uppercase letters.	Application launches successfully.	
TC15	Launch an application with lowercase letters.	Application launches successfully.	
TC16	Launch an application with mixed-case letters.	Application launches successfully.	
TC17	Launch an application with leading/trailing spaces in the command.	Extra spaces are ignored and application opens.	
TC18	Launch an application that is not installed.	"Application not found" message is displayed.	
TC19	Launch an application with an empty name.	Validation message is displayed.	
TC20	Launch an application using only spaces.	Launch request is rejected.	
TC21	Launch multiple different applications one after another.	All applications open successfully.	
TC22	Launch the same application multiple times.	Application opens each time or existing instance is activated.	
TC23	Launch an application immediately after system startup.	Application opens successfully.	

TC24	Launch an application after restarting NOVA.	Application opens successfully.	
TC25	Launch an application while another application is already running.	Requested application opens successfully.	
TC26	Close the launched application manually.	Application closes successfully.	
TC27	Launch an application after closing it.	Application opens again successfully.	
TC28	Launch an application using a voice command.	Correct application opens.	
TC29	Receive application name from Command Processing.	Correct application name is received.	
TC30	Verify Registry Launcher provides the executable path.	Valid executable path is received.	
TC31	Launch application using the executable path received from Registry Launcher.	Application launches successfully.	
TC32	Registry Launcher returns an invalid path.	Appropriate error message is displayed.	
TC33	Registry Launcher does not find the application.	Launch operation is cancelled with an error message.	
TC34	Verify no hard-coded application path is used.	Application launches through Registry information only.	
TC35	Launch application when Windows permissions are restricted.	Appropriate permission error is displayed.	
TC36	Launch application after Windows user login.	Application opens successfully.	

TC37	Launch application while CPU usage is high.	Application launches without crashing.	
TC38	Launch application while memory usage is high.	Application launches successfully.	
TC39	Perform 20 application launches continuously.	All launch requests are processed successfully.	
TC40	Measure application launch response time.	Application launches within acceptable time.	
TC41	Verify application launch logs are created.	Launch activity is logged successfully.	
TC42	Restart the Application Launcher module.	Module restarts successfully.	
TC43	Launch an application after restarting the module.	Application opens successfully.	
TC44	Close the Application Launcher while a launch request is in progress.	Module closes safely without crashing.	
TC45	Launch application after an unexpected module restart.	Launch works normally.	
TC46	Verify compatibility on Windows 10.	Module functions correctly.	
TC47	Verify compatibility on Windows 11.	Module functions correctly.	
TC48	Verify launched application receives focus on the desktop.	Opened application becomes active.	
TC49	Exit the Application Launcher after completing operations.	Module exits successfully.	
TC50	Close the NOVA application after launching applications.	NOVA closes without affecting launched applications.	

Module: Command Processing :

Functional Requirements

- Receive recognized text from Voice Recognition.
- Identify the user's intent.
- Forward the command to the correct module.
- Handle invalid or unsupported commands.
- Return the response to Text-to-Speech.

TC ID	Test Case Description	Expected Result	Result (Pass/Fail)
TC01	Receive text from the Voice Recognition module.	Text is received successfully.	
TC02	Receive the command "Open Notepad".	Command is recognized correctly.	
TC03	Receive the command "Open Calculator".	Command is recognized correctly.	
TC04	Receive the command "Open Chrome".	Command is recognized correctly.	
TC05	Receive the command "Open Settings".	Settings command is identified.	
TC06	Receive the command "Create Folder Test".	File Management command is identified.	
TC07	Receive the command "Search File Resume.pdf".	Search command is identified.	
TC08	Receive the command "Enable Startup".	Windows Startup command is identified.	
TC09	Receive the command "Disable Startup".	Windows Startup command is identified.	
TC10	Receive the command "Launch Paint".	Application Launcher command is identified.	

TC11	Identify the user's intent for opening an application.	Intent is detected correctly.	
TC12	Identify the user's intent for file operations.	Correct module is selected.	
TC13	Identify the user's intent for settings changes.	Settings module is selected.	
TC14	Identify the user's intent for startup operations.	Startup module is selected.	
TC15	Receive an unsupported command.	"Command not supported" message is returned.	
TC16	Receive random text instead of a command.	Invalid command message is displayed.	
TC17	Receive an empty command.	Validation message is displayed.	
TC18	Receive a command containing only spaces.	Command is ignored.	
TC19	Receive a command with extra spaces.	Extra spaces are ignored.	
TC20	Receive a command with uppercase letters.	Command is processed successfully.	
TC21	Receive a command with lowercase letters.	Command is processed successfully.	
TC22	Receive a command with mixed-case letters.	Command is processed successfully.	
TC23	Receive a command containing numbers.	Numbers are processed correctly.	
TC24	Receive a command containing special characters.	Invalid characters are handled correctly.	

TC25	Forward an application command to the Application Launcher.	Command reaches the Application Launcher.	
TC26	Forward a Registry search command to the Registry Launcher.	Command reaches the Registry Launcher.	
TC27	Forward a File operation command to File Management.	Command reaches File Management.	
TC28	Forward a Startup command to Windows Startup.	Command reaches Startup module.	
TC29	Forward a Settings command to Settings Management.	Command reaches Settings module.	
TC30	Forward a valid response to Text-to-Speech.	Response is sent successfully.	
TC31	Return an error response for an invalid command.	Error response is sent to Text-to-Speech.	
TC32	Process multiple commands one after another.	Every command is processed correctly.	
TC33	Process duplicate commands.	Every command is handled correctly.	
TC34	Receive a very long command.	Complete command is processed.	
TC35	Receive a one-word command.	Command is processed successfully.	
TC36	Restart the Command Processing module.	Module restarts successfully.	
TC37	Process commands after restarting the module.	Commands are processed correctly.	
TC38	Process commands continuously for 30 minutes.	No crash occurs.	

TC39	Process 100 consecutive commands.	All commands are processed successfully.	
TC40	Verify command processing response time.	Response is within acceptable time.	
TC41	Verify command logs are generated.	Logs are created successfully.	
TC42	Process commands while CPU usage is high.	Module remains responsive.	
TC43	Process commands while memory usage is high.	Commands are processed correctly.	
TC44	Close the module during command execution.	Module closes safely.	
TC45	Restart after an unexpected shutdown.	Module works correctly.	
TC46	Verify compatibility on Windows 10.	Module works successfully.	
TC47	Verify compatibility on Windows 11.	Module works successfully.	
TC48	Exit the Command Processing module.	Module exits successfully.	
TC49	Verify no command is lost during processing.	All commands are processed correctly.	
TC50	Verify system remains stable after continuous command processing.	No unexpected errors occur.	

Module: Sleep/Wake Mode (TC01–TC50)

Functional Requirements

- Enter sleep mode on user command.
- Wake up using a wake word (e.g., "NOVA").
- Ignore normal commands while sleeping.
- Resume listening after wake-up.
- Continue running in the background.
- Handle invalid wake commands gracefully.
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TC ID	Test Case Description	Expected Result	Result (Pass/Fail)
TC01	Start the Sleep/Wake Mode module.	Module starts successfully.	
TC02	Give the command "Go to Sleep".	NOVA enters Sleep Mode.	
TC03	Verify the assistant stops listening after entering Sleep Mode.	Assistant stops processing normal commands.	
TC04	Give the wake word "NOVA".	Assistant wakes up successfully.	
TC05	Give an incorrect wake word.	Assistant remains in Sleep Mode.	
TC06	Speak a normal command while the assistant is sleeping.	Command is ignored.	
TC07	Repeat the wake word twice.	Assistant wakes only once.	
TC08	Speak the wake word loudly.	Assistant wakes successfully.	

TC09	Speak the wake word softly.	Assistant wakes if the voice is clear.	
TC10	Speak the wake word with background noise.	Assistant wakes or asks the user to repeat.	
TC11	Stay silent after entering Sleep Mode.	Assistant remains in Sleep Mode.	
TC12	Wake the assistant after 5 minutes of Sleep Mode.	Assistant wakes successfully.	
TC13	Wake the assistant after 30 minutes of Sleep Mode.	Assistant wakes successfully.	
TC14	Wake the assistant after system idle time.	Assistant resumes listening.	
TC15	Verify the assistant displays a wake-up message.	Wake-up confirmation is displayed or spoken.	
TC16	Verify the assistant displays a sleep confirmation message.	Sleep confirmation is displayed or spoken.	
TC17	Put the assistant into Sleep Mode multiple times.	Sleep Mode activates correctly every time.	
TC18	Wake the assistant multiple times.	Assistant wakes correctly every time.	
TC19	Give the Sleep command while already in Sleep Mode.	No duplicate action occurs.	
TC20	Give the Wake command while already awake.	No duplicate action occurs.	
TC21	Restart the application while in Sleep Mode.	Assistant starts in the default mode.	
TC22	Restart the application while awake.	Assistant starts normally.	

TC23	Minimize the application during Sleep Mode.	Sleep Mode continues in the background.	
TC24	Restore the application from minimized state.	Previous mode is maintained.	
TC25	Lock the Windows system during Sleep Mode.	Sleep Mode continues correctly.	
TC26	Unlock Windows and give the wake word.	Assistant wakes successfully.	
TC27	Put the system into Sleep Mode and resume.	Assistant continues functioning correctly.	
TC28	Hibernate the system and resume.	Assistant resumes correctly.	
TC29	Verify the assistant runs in the background during Sleep Mode.	Background service remains active.	
TC30	Verify CPU usage during Sleep Mode.	CPU usage remains low.	
TC31	Verify memory usage during Sleep Mode.	Memory usage remains stable.	
TC32	Give multiple normal commands while sleeping.	All commands are ignored.	
TC33	Give multiple wake words continuously.	Assistant wakes only once.	
TC34	Wake the assistant and immediately give a command.	Command is processed successfully.	
TC35	Verify Voice Recognition resumes after wake-up.	Voice Recognition starts automatically.	
TC36	Verify Command Processing resumes after wake-up.	Commands are processed successfully.	

TC37	Verify Text-to-Speech responds after wake-up.	Assistant responds correctly.	
TC38	Disconnect the microphone during Sleep Mode.	Appropriate error message is displayed.	
TC39	Reconnect the microphone and give the wake word.	Assistant wakes successfully.	
TC40	Verify the assistant does not crash during Sleep/Wake transitions.	No crash occurs.	
TC41	Perform 50 Sleep/Wake cycles continuously.	Module remains stable.	
TC42	Verify response time for entering Sleep Mode.	Sleep Mode activates within acceptable time.	
TC43	Verify response time for waking up.	Assistant wakes within acceptable time.	
TC44	Verify logs are generated for Sleep Mode.	Sleep event is logged successfully.	
TC45	Verify logs are generated for Wake Mode.	Wake event is logged successfully.	
TC46	Close the application while in Sleep Mode.	Application closes safely.	
TC47	Restart the Sleep/Wake module.	Module starts successfully.	
TC48	Verify compatibility on Windows 10.	Module works correctly.	
TC49	Verify compatibility on Windows 11.	Module works correctly.	
TC50	Exit the application after Sleep/Wake operations.	Application exits without errors.	